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Supplementary Materials for

All-optical synthesis chip for large-scale intelligent semantic vision generation

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Other Supplementary Material for this manuscript includes the following:

Movie S1

Materials and Methods

Dimensions and further validations of OLS

The influence of the dimension of OLS is shown in Fig. S11. The loss decreased and the appearance quality of the output improved as the dimension of OLS increased until 100 (10×10). When the dimension of OLS was adequate, dimensionality increasing did not bring prominent improvements. Sometimes it might even enlarge the difficulty to train the weight to a global optimum (Fig. S11A-B).

We also experimentally compared the OLS function with the latent spaces in digital VAE (*25*) and β-VAE (*40*) (Fig. S12). Via OLS, LightGen experimentally manipulated the semantic features and varied the generated objects with continuously changed features and perspectives, showing that OLS had functionalities comparable to latent spaces in digital VAE and β-VAE.

Loss function of BOGT

We used Kullback-Leibler divergence (KL divergence) (*26*) to constrain $Q(Z|X)$ close enough to $P(Z)$ during training with BOGT. According to the fact that the intensity of light is always positive, we used a truncated biased Gaussian distribution, unlike the conventional normal Gaussian, to fit the peculiarities of LightGen. Then the KL divergence became $D(Q(Z|X) \parallel P(Z)) = \frac{1}{2}[\mathrm{tr}(\sum^2) + (M-M_0)^{\mathrm{T}}(M-M_0) - k - \log(\det(M^2))]$, where k was the dimension of OLS; M and M_0 were the means of Q and P respectively. The loss function of LightGen for training was then defined as $L = \alpha l_{KLD} + \beta l_{mse} + \gamma l_{op}$, where l_{KLD} was the KL divergence to guarantee the estimated distribution in OLS was close enough to the goal; l_{mse} was an alternative mean square error of the LightGen output from input; l_{op} was a penalty term to prevent the light field between layers from being too sick or too dim for physical implementation, like some deep neural networks usually do; α, β and γ were the coefficients to weight each loss.

Experimental set-up of LightGen chip

The sketch and experimental set-up for LightGen are shown in Fig. S13. We used a single-mode 532nm laser (Changchun New Industries Optoelectronics Tech. Co., Ltd, MSL-U-532-300mW) to generate collimating incident light. A secondary cascaded beam expander system was used to expand the diameter of the beam. We used an amplitude-modulation spatial light modulator (HOLOEYE Photonics AG, HES6001) to generate the input. A 4f relay system was placed between the input and LightGen chip to conjugate the input image onto the LightGen input plane. We used an sCMOS (Tucsen Photonics Co., Ltd, Dhyana 400BSI) to measure the intensity output from LightGen. Because the detection distance is 3 mm, a lens was also used between the chip and the camera to image on the designed detection plane. Before fabrication, separate validation of a single layer could also be done with a phase-only spatial light modulator (Meadowlark Optics, Inc. P1920-400-800-PCIE) to facilitate the reconfiguration.

Chip design of LightGen

The demonstration of LightGen in the manuscript was composed of a two-layer encoder and a four-layer generator (Fig. 1F-I), with 400×400 or 600×600 photonic neurons in each layer, depending on the task. The three-channel tasks were implemented with three independent R, G, B channels. For tasks that

adjust the OLS to generate, we only used the photonic generator. Each photonic neuron had a pixel size of 3 μm × 3 μm, 8 levels, fabricated with overlay photolithography on JGS1 (Fig. S14). JGS1 is a silicon dioxide material with a high deformation point (1120 °C) and robust mechanical properties (with a design bending strength >60 MPa). Its thermal expansion coefficient is 5.5×10^{-7} cm/cm·°C (*45*). It means that even if the temperature change reaches 100 °C, which is significantly higher than usual thermal drifts in this chip, it only results in a deformation of 0.006% of the overall size of the chip. This error is orders of magnitude lower than the threshold (10%) to cause output degradation.

The thickness difference between each level was 150nm. The overall thickness of the JGS1 was 0.6 mm. The detailed unit cell schematic for the 3μm pixels in the meta-surfaces is shown in Fig. S15A. Each pixel had different depths to implement different weights. Because of the difference in refractive index between air and SiO_2, we etched different depths in the meta-surfaces, resulting in diverse optical path differences, therefore different phases of the wavefront. So we modulated the diffraction and interference of the light via meta-surfaces with pixels of different depths. This is a widely accepted way to modulate the light field in photonic computing (*10,23,46*). The phase corresponding to the depth was derived from:

$$\phi(d)=\frac{2\pi d}{\lambda}(n-n_0) \tag{S1}$$

where n, n_0 is the reflective index of SiO_2 and air; λ is the wavelength of the light in air; d is the depth of the pixel; $\phi(d)$ is the phase difference corresponding to the depth d. Due to the limitations of the photolithography process, the weights were quantized into 8 levels and etched onto the SiO_2 layers (Fig. S15B). Therefore, the corresponding phase weights of step 0, 1, 2, 3, 4, 5, 6 and 7 were 0 rad, 0.82 rad, 1.63 rad, 2.45 rad, 3.26 rad, 4.08 rad, 4.89 rad and 5.71 rad, respectively, at 532 nm. The mapping relationship of modulation phases and the steps are shown in Fig. S15B. After quantizing the trained phase into discrete 8 levels, LightGen preserved task performance. We also displayed the high-resolution SEM and AFM images of meta-surfaces in Fig. S16A-B. The measured SEM images of meta-surfaces showed the high-resolution view of pixels with diverse depths. The distribution and pixel size corresponded with the design. The measured AFM photos of zoomed-in parts showed the actual etch depth and detailed features of the 3μm pixels also corresponded well with the design.

The OLS was implemented with a 10×10 single-mode fiber array, with fiber cores and cladding. Its length is 8.5 mm and the external diameter of each fiber is 125 μm. The measured microscopic image of the fiber array is shown in Fig. 1H. The meta-surfaces and the fiber array were packaged and fixed onto a custom-designed 3D printed holder (Fig. S17). The top and bottom holders, i.e., the substrate and lid, were equipped with corresponding grooves to align and fix the meta-surfaces and fiber array. We aligned them with perpendicular microscopies and packaged them with adhesive.

Quantization of the implemented weights of LightGen

We display our detailed quantization method in Fig. S18A. Due to the limitations of the photolithography process, the weights were quantized into 8 levels and etched onto the SiO_2 meta-surface layers and the depth difference between each level was 150nm. Therefore, the corresponding phase weights of step 0, 1, 2, 3, 4, 5, 6 and 7 were 0 rad, 0.82 rad, 1.63 rad, 2.45 rad, 3.26 rad, 4.08 rad, 4.89 rad and 5.71 rad, respectively, at 532 nm. The mapping relationship of modulation phases and the

steps is shown in Fig. S18A. We quantized the high-precision phase weights into their nearest quantized phase weights among 0, 0.82, 1.63, 2.45, 3.26, 4.08, 4.89 and 5.71 rad. An example of the phase weights before and after quantization in the meta-surface for 3D generation task is shown in Fig. S18B. We also zoomed in on the framed parts in Fig. S18C. The 8-level quantization well retained the trends of high-precision masks.

We further displayed both the simulation output with (W/) and without (W/O) 8-level quantization of all six layers, and the experimental output W/ quantization of all six layers of LightGen (Fig. S18D). The similarity between the simulation W/ and W/O quantization showed that 8 levels were adequate to generate expected outputs (Fig. S18D, the first and second row). The experimental output by LightGen further demonstrated the maintained experimental performance after quantization of all six layers (Fig. S18D, the third row). The feasibility of quantizing the phase weights and maintaining performances had also been validated by (*14*).

Packaging and calibration of LightGen

The layers were packaged with perpendicular microscopies onto a 3D printed holder with a substrate and a lid, with printing error lower than 2 μm, constraining the layers to align precisely. Because the grooves were precisely aligned, as long as the meta-surfaces are precisely fixed onto the holder, the encoder, OLS and the generator were automatically aligned (Fig. S19A). For some extraordinary cases that the meta-surfaces exhibited misalignment intra-layers or inter-layers, we could also calibrate them as demonstrated in Fig. S19B. The intra-layer misalignment means the misalignment between each overlay lithography etching layer. According to the SEM and AFM photos measured with the meta-surfaces in LightGen in Fig. S16, the intra-layer misalignment was commonly below 600 nm, i.e., below 20% pixel size as the pixel size of each optical neuron on the meta-surfaces was 3.0 μm. We showed the PSNR of the generated images by LightGen after misalignment of 10%, 30% and 50% pixel in etching. The misalignment of overlay etching had little interference to the final output in both quantitative (Fig. S19B, the green lines) and qualitative aspects (Fig. S19C), even under misalignment of 50% pixel, which was much higher than its common etching error (below 20% pixel). Additionally, we could retrain and replace the following layers to further correct the decreased PSNR from 13.80 dB back to 16.05 dB. The mask weights of misalignment and after retraining are also shown in Fig. S19E.

For the inter-layer misalignment, we showed the influence of misalignment of 10%, 50% and 100% pixel between meta-surfaces. The 3D printing error was below 2 μm, so the misalignment was commonly below 67% pixel. Small misalignment resulted in PSNR still almost 15 dB and good perceptual quality (Fig. S19B, yellow lines and Fig. S19D). Whereas if the misalignment exceeded 50% pixel, we could also retrain and replace the meta-surfaces in the following grooves. The qualitative and quantitative output could be effectively corrected back after calibration and retraining (Fig. S19B, D and F).

Benefiting from the 3D packaging of LightGen, six meta-surfaces with 2.16 million photonic neurons could be integrated into an area of 136.5 mm^2. Because it combined both free-space components and waveguide components in a 3D space, we named it 3D packaging. We further listed a comparison of packaging methods with existing photonic computing systems (Table S2). It showed that the key innovation of 3D packaging lay in overcoming the dilemma where MZI- and MRR-based devices

usually had limited integration scales, whereas diffractive networks with large-scale neurons usually required desktop systems with large volumes. LightGen achieved large-scale photonic neurons and integration simultaneously.

Training style-transfer and denoising LightGen

For the training of style-transfer LightGen, we used pre-trained multi-layer weights to extract the content and style features for the calculation of loss function (*32*). The top layers extracted high-level content of the image but did not impose tight constraints on detailed pixels. The bottom layers focused more on the precise pixel values from the image. Therefore, we used the top layers to capture the image content representation (*47*). The content loss ($l_{Content}$) was given by the mean squared error between the two content representations from the LightGen generated images and the input content images. Meanwhile the style loss (l_{Style}) was obtained by calculating the mean-squared error between the Gram matrix output from the images generated by the LightGen and the style images (*47*). The total loss was given by $l_{Total} = l_{Style} + \alpha \times l_{Content}$, where α was the ratio of the content loss. The network was trained with Adam optimizer (*48*). With the balancing of $l_{Content}$ and l_{Style}, LightGen experimentally generated style-transferred images with both target styles and original contents, as shown in Fig. 3E-F and Fig. S20. For the patched style transfer method for comparison in Fig. 3G, we trained the network with the same style-transfer architecture but input in 7 × 7 patches to simulate the influence of limited input dimensions. For example, for a 28 × 28 image in EMNIST dataset, it was input in 16 separate 7 × 7 patches. It turned out to have discontinuous boundaries after normalized assembly (Fig. 3G).

It was also optional to add simple processes after generation with LightGen in colorful generation. As shown in Fig. 3A, after generation with LightGen, we extracted the brightness (the 'value' channel of HSV representation) as I_1 and the results of median filtering applied to I_1 after normalization as I_2. Then we applied histogram matching to I_2 to calibrate the color as I_3. The images in Fig. 3A were represented as the weighted sum of I_2 and I_3. The post-processing was optional, which was not applied to other results in the figures.

For the LightGen designed for denoising, we trained LightGen with the l_{mse} in the loss function calculated with the ground truth without noise and the output of LightGen, instead of calculating with the noisy input and LightGen output, to ensure the denoising performance of LightGen. During the training, LightGen learned to remove noise from the input images and retain semantic features, which eventually enabled it to be robust to random noise. It was trained via Adam optimizer, with 400×400 trainable pixels in each layer. We used the Kuzushiji-Modified National Institute of Standards and Technology (K-MNIST) dataset for demonstration. The denoising capability of LightGen was experimentally validated under four levels of noise with standard deviation values of 0.2, 0.3, 0.4 and 0.5, respectively. The experimental results of denoising are shown in Fig. 3C-D. LightGen achieved exceeding denoising performance in both quantitative and qualitative results.

Modeling of wave propagation among meta-surfaces

According to the Rayleigh-Sommerfeld diffraction law (*49, 50*), each trainable neuron in the meta-surface can be viewed as a secondary wave source. Its optical field distribution can be written as

$$w_i^l(x,y,z)=\frac{z-z_i}{r^2}\left(\frac{1}{2\pi r}+\frac{1}{j\lambda}\right)\exp\left(\frac{j2\pi r}{\lambda}\right) \tag{S2}$$

Where (x, y, z) is the coordinates of the point of interest; λ is the wavelength of the coherent incident light; the index l is the l-th layer of the meta-surfaces; j is the imaginary unit $\sqrt{-1}$. The index i refers to the i-th diffractive neuron in this layer, whose coordinates are (x_i, y_i, z_i). The distance r from the i-th diffractive neuron to the interested point (x, y, z) in the optical field can be written as

$$r=\sqrt{(x-x_i)^2+(y-y_i)^2+(z-z_i)^2} \tag{S3}$$

Hence, we obtained the phase–amplitude coupling relationship dependent on the input light wave and the neurons in the meta-surfaces. Therefore, the output at (x, y, z) by the i-th neuron at (x_i, y_i, z_i) in an l-layer diffractive neural network can be obtained as:

$$n_i^l(x,y,z)=w_i^l(x,y,z)t_i^l(x_i,y_i,z_i)\sum_k n_k^{l-1}(x_i,y_i,z_i)=w_i^l(x,y,z)|A|e^{j\Delta\theta} \tag{S4}$$

where $|A|$ is the relative amplitude of the secondary wave; $\sum_k n_k^{l-1}(x_i,y_i,z_i)$ is the input light to the i-th neuron in the l-th layer; $\Delta\theta$ is the phase change introduced to the secondary wave after modulation by the diffractive neuron, which depends on the transmission coefficient. In this way, the secondary waves propagate and diffract between and beyond the diffractive layers (meta-surfaces), interfering with one another. The transmission coefficient is a complex variable comprising both amplitude and phase terms. It can be written as:

$$t_i^l(x_i,y_i,z_i)=A_i^l(x_i,y_i,z_i)e^{j\varphi_i^l(x_i,y_i,z_i)} \tag{S5}$$

where the amplitude modulation and phase modulation constitute the two constituent parts. Here in meta-surfaces of LightGen, we mainly modulated the phases with a constant transmission intensity loss. In this way, we modelled the wave propagation and the mutual interactions among the meta-surface layers in LightGen. The diffractive meta-surfaces therefore established a neural network in which each layer performed complex-valued operations.

Modeling and coupling of OLS

The fundamental mode of the single-mode fiber (approximately Gaussian) is given as:

$$E_{\text{fiber}}(x,y)=E_0\exp\left(-\frac{x^2+y^2}{w_f^2}\right) \tag{S6}$$

where E_0 is the peak electric field amplitude at the fiber center; w_f is the mode field radius; x and y are the transverse coordinates.

To normalize the total power P of the mode:

$$P=\iint_{-\infty}^{\infty}|E_{\text{fiber}}(x,y)|^2\,dxdy=1 \tag{S7}$$

Using equation (S6), we obtained:

$$\left|E_{\text{fiber}}(x,y)\right|^2 = \left|E_0\right|^2 \exp\left(-\frac{2\left(x^2+y^2\right)}{w_f^2}\right) \tag{S8}$$

We got the integral after calculating in polar coordinates:

$$P = |E_0|^2 \cdot 2\pi \cdot \frac{w_f^2}{4} \tag{S9}$$

Then:

$$|E_0|^2 = \frac{2}{\pi w_f^2} \tag{S10}$$

$$E_0 = \sqrt{\frac{2}{\pi w_f^2}} \tag{S11}$$

Therefore, the normalized fiber mode field became:

$$E_{fiber}(x,y) = \sqrt{\frac{2}{\pi w_f^2}} \exp\left(-\frac{x^2+y^2}{w_f^2}\right) \tag{S12}$$

The input electric field E_{in} was a uniform plane wave over an $L \times L$ region, where L was the length of the plane wave source, measured in micrometers (μm):

$$E_{in}(x,y) = \begin{cases} \frac{1}{L}, |x| \le \frac{L}{2}, |y| \le \frac{L}{2} \\ 0, otherwise \end{cases} \tag{S13}$$

after normalization $\iint_{-\infty}^{\infty} |E_{in}(x,y)|^2 \, dxdy = 1$.

The coupling efficiency η was defined as the overlap integral squared between the input field and the fiber mode (*51*):

$$\eta = \left|\iint E_{in}(x,y) E_{fiber}^*(x,y) dxdy\right|^2 \tag{S14}$$

Since $E_{\text{fiber}}^* = E_{\text{fiber}}$, from equation (S14) we obtained the expressions for the fields:

$$\eta = \left(\frac{1}{L}\sqrt{\frac{2}{\pi w_f^2}} \iint_{-\frac{L}{2}}^{\frac{L}{2}} \exp\left(-\frac{x^2+y^2}{w_f^2}\right) dxdy\right)^2 \tag{S15}$$

By extracting the symmetric portion of the integral from the equation (S14) and treating x and y equivalently, we separated the variables:

$$\iint_{-\frac{L}{2}}^{\frac{L}{2}} \exp\left(-\frac{x^2+y^2}{w_f^2}\right) dxdy = \left(\int_{-\frac{L}{2}}^{\frac{L}{2}} \exp\left(-\frac{x^2}{w_f^2}\right) dx\right)^2 \tag{S16}$$

For the single-variable integral I:

$$I = \int_{-L/2}^{L/2} \exp\left(-\frac{x^2}{w_f^2}\right) dx \tag{S17}$$

Let $u=x/w_f$, the result was:

$$I = w_f \int_{-L/(2w_f)}^{L/(2w_f)} e^{-u^2} du = w_f \sqrt{\pi}\,\mathrm{erf}\left(\frac{L}{2w_f}\right) \tag{S18}$$

where erf was the error function, defined as $\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2}\, dt$.

Thus, the double integral became:

$$\iint_{-\frac{L}{2}}^{\frac{L}{2}} \exp\left(-\frac{x^2+y^2}{w_f^2}\right) dxdy = w_f^2 \pi \left[\mathrm{erf}\left(\frac{L}{2w_f}\right)\right]^2 \tag{S19}$$

Substituting this expression into the coupling efficiency formula gave:

$$\eta = \left(\frac{1}{L}\sqrt{\frac{2}{\pi w_f^2}} \cdot w_f^2 \pi \left[\mathrm{erf}\left(\frac{L}{2w_f}\right)\right]^2\right)^2 \tag{S20}$$

After Simplification:

$$\eta = \frac{2\pi w_f^2}{L^2}\left[\mathrm{erf}\left(\frac{L}{2w_f}\right)\right]^4 \tag{S21}$$

When considering various incident angles, it was equivalent to the incident electric field with a rotated phase in direction that we defined as x. The input field then was:

$$E_{\mathrm{in}}(x,y) = \begin{cases} \frac{1}{L} exp(iK_x x), |x| \le \frac{L}{2}, |y| \le \frac{L}{2} \\ 0, else \end{cases} \tag{S22}$$

where $K_x = k\sin(\theta), k = \frac{2\pi}{\lambda}$, and θ is the incidence angle, λ is the wavelength of the light, K_x is the wave vector component in the x-direction.

The normalized fiber mode field was equation (S12) and the coupling efficiency was defined as the overlap integral squared between the input field and the fiber mode as equation (S14).

Using Equations (S12) and (S22) in Equation (S14), we derived:

$$\eta = \left| \iint_{-L/2}^{L/2} \frac{1}{L} \exp(iK_x x) \sqrt{\frac{2}{\pi w_f^2}} \exp\left(-\frac{x^2+y^2}{w_f^2}\right) dxdy \right|^2 \tag{S23}$$

Focusing on the integral portion of the above formula in terms of x and y, we obtained:

$$\begin{aligned} \iint_{-\frac{L}{2}}^{\frac{L}{2}} \exp(iK_x x) \exp\left(-\frac{x^2+y^2}{w_f^2}\right) &= \int_{-\frac{L}{2}}^{\frac{L}{2}} \exp\left(-\frac{x^2}{w_f^2}\right) \exp(iK_x x) \exp\left(-\frac{y^2}{w_f^2}\right) dxdy \\ &= \left(\int_{-\frac{L}{2}}^{\frac{L}{2}} \exp\left(-\frac{x^2}{w_f^2}\right) \exp(iK_x x) dx\right)\left(\int_{-\frac{L}{2}}^{\frac{L}{2}} \exp\left(-\frac{y^2}{w_f^2}\right) dy\right) \end{aligned} \tag{S24}$$

The result for the y-dependent part of the integral was given in equation (S18), then the integration in the x-direction was defined as J_x:

$$J_x = \int_{-L/2}^{L/2} \exp\left(-\frac{x^2}{w_f^2}\right) \exp\left(iK_x x\right) dx \tag{S25}$$

The result of the integration was:

$$J_x = w_f \sqrt{\pi} \exp\left(-\frac{K_x^2 w_f^2}{4}\right) \mathrm{Re}\left(\mathrm{erf}\left(\frac{L}{2w_f} + i\frac{K_x w_f}{2}\right)\right) \tag{S26}$$

Combining the integrations in the *x* and *y*-directions, the final result was:

$$\eta = \frac{2\pi {w_f}^2}{L^2}\left[\mathrm{erf}\left(\frac{L}{2w_f}\right)\right]^2 \exp\left(-\frac{K_x^2 {w_f}^2}{2}\right)\left[\mathrm{Re}\left(\mathrm{erf}\left(\frac{L}{2w_f} + i\frac{K_x w_f}{2}\right)\right)\right]^2 \tag{S27}$$

Finally, the derived formula was evaluated at a series of incident angles: 0°, 2.5°, 5°, 7.5°, 10°, 20°, 40°, 60°, and 80°. As shown in Fig. S2A, the theoretical and simulated coupling efficiency corresponded precisely under various incident angles. Figure S2B-D shows the input and output light fields that propagated through OLS in LightGen based on the above modeling of complex coupling.

Cross-validation of experimental computing metrics of LightGen

We compared the end-to-end systemic time, energy consumption and computing density of LightGen, with state-of-the-art electronic chips and algorithms, and photonic chips implementing generative tasks (Table S1). We noted that for continuous linear diffractive layers without nonlinear activations, we calculated them as one equivalent single layer with reduced operation numbers in the evaluation of LightGen for fair comparison. Under this more conservative calculation, LightGen remained to experimentally achieve 3.57×10^4 TOPS, 6.64×10^2 TOPS/W and 2.62×10^2 TOPS/mm^2 in systemic computing speed, energy efficiency and computing density, respectively, more than 2, 2 and 2 orders of magnitude higher than NVIDIA A100 (Table S1 and Supplementary Text 1). With an ultrafast spatial light modulator (SLM) that is not limited by the refresh rate for input, LightGen could further achieve a theoretical computing speed of 5.69×10^9 TOPS (Supplementary Text 1 and Table S1).

To further validate the performance of LightGen, we used a stricter metric to measure the end-to-end time and energy consumption of LightGen when it not only completed the task but also achieved comparable performance with state-of-the-art electronic models, to ensure the practical application of LightGen. Figure S9 shows the end-to-end experimental average time and energy consumption of LightGen and StyleGAN on NVIDIA A100 per frame. When achieving comparable style transfer indices (Fig. 3F and Fig. S6C-D), LightGen experimentally consumed 0.002 s/frame and 0.113 J/frame (Fig. S9, the orange bar). And the forward inference for per frame by StyleGAN (codes from (*52*)) on NVIDIA A100 consumed 0.04 s/frame and 4.36 J/frame as measured (Fig. S9, the blue bar). The measured time corresponded well with published reports (*53*). LightGen experimentally reduced the end-to-end computing time and energy consumption by orders of magnitude, aligning well with the modeling results in Table S1 (the ninth row, experimental LightGen). When using a faster SLM for input, LightGen further could potentially achieve 0.09 ns/frame and 4.90 nJ/frame theoretically (Fig. S9, the yellow bar), which also aligns well with the modeling results in Table S1 (the tenth row, theoretical LightGen). More details of measurement and calculation are included in Supplementary Text 1.

Supplementary Text 1: Measurement and calculation of the systematic computing speed, computing density and energy efficiency of LightGen

The encoder and generator of LightGen were composed of two and four layers with 400 × 400 or 600 × 600 pixels each layer (Table S3). The pixel number depended on the tasks. For clearer narration, we took 600 × 600 pixels each layer as an example, but we listed both in Table S3 to facilitate validation.

For continuous linear diffractive layers without nonlinear activations, instead of calculating them as multiple layers with multiplicative operation numbers, we calculated them as one equivalent single layer with reduced operation numbers in the evaluation of LightGen here for fair comparison. So the operation number was calculated with the neuron numbers of input and output to be 5.18×10^{11} for a single forward inference. The computing time was dependent on the propagation distance. The length of the 3D-printed holder to package all components of LightGen was 27.3 mm (Table S4). Therefore the theoretical computing time was 9.1×10^{-11} s.

The computing speed S was calculated with the formula

$$\text{Computing speed } S=\frac{N_{op}}{t_{frame}}, \tag{S28}$$

where N_{op} represented the total operation number implemented for one frame, i.e., single forward inference; t_{frame} was the time for LightGen to end-to-end process one frame. Therefore, the end-to-end theoretical computing speed for LightGen was 5.69×10^{9} TOPS.

In experiments, we used an amplitude-modulation SLM (HOLOEYE Photonics, HES6001), with an input frame rate of 60 Hz with parallel processing. The resolution of the SLM is 1920×1080. To generate style transferred or denoised images such as in Fig. 3D-E, we required to input the original images (including paddings to eliminate crosstalk) with SLM as the input of LightGen. Considering adequate intervals between inputs, we could reuse each SLM frame as 4×2 inputs for LightGen. Therefore, considering the experimental latency of SLM, LightGen had a frame rate of 4×2×60 Hz = 480 Hz. The operation number of LightGen for using the all-optical encoder, OLS and generator for semantic generation was $2\times(2\times600^4 - 600^2) = 5.18\times10^{11}$. Therefore, the end-to-end experimental computing speed in serial of LightGen was $5.18\times10^{11}\times480$ Hz = 2.49×10^{2} TOPS. To generate semantic-manipulated media such as in Fig. 3A and 4C-F, we only needed to input a 10×10 image to represent the optical latent values to the generator. Considering adequate intervals between inputs, we reused each SLM frame as 64×36 inputs for LightGen. Therefore, considering the experimental latency of SLM, LightGen had a frame rate of 64×36×60 Hz = 138 kHz. The operation number of LightGen for using the all-optical generator for semantic generation was $2\times600^4 - 600^2 = 2.59\times10^{11}$. Therefore, the end-to-end experimental computing speed in serial of LightGen was $2.59\times10^{11}\times138$ kHz = 3.57×10^{4} TOPS. It showed that even for the systemic computing speed with not top-tier SLM, LightGen experimentally achieved 2 orders of magnitude improvement over NVIDIA A100.

The computing density D was calculated as:

$$\text{Computing density } D=\frac{S}{A}, \tag{S29}$$

where A is the footprint of the chip. The footprints of all components in LightGen are shown separately in Table S4. All components were packaged within the 3D-printed holder, composed of a lid and

substrate with the same footprints. The integrated LightGen, sharing the footprint of the 3D-printed holder, had a length and width of 27.3 mm and 5 mm, respectively (Table S4). So the footprint of integrated LightGen was 136.5 mm^2. Then the theoretical computing density of LightGen was 5.69×10^9 /136.5 = 4.17×10^7 TOPS/mm^2. The experimental computing density of LightGen was $3.57\times10^4/136.5 = 2.62\times10^2$ TOPS/mm^2.

The energy consumption of LightGen was composed of the energy of incident coherent light, spatial light modulator to modulate the incident light and the camera to detect the output. Their measured power values were 249.3 μW, 4.56 W and 49.29 W, respectively. The total power consumption of LightGen was thus 53.85W. Therefore the theoretical energy consumption of LightGen was 4.90 nJ per frame.

The systematic energy efficiency was calculated as:

$$\text{Energy efficiency}_{sys} = \frac{S}{E_{sys}}, \tag{S30}$$

where the E_{sys} was the systematic energy consumption of LightGen for one frame. Therefore, the theoretical systematic energy efficiency of LightGen was 1.06×10^8 TOPS/W. The experimental systematic energy efficiency of LightGen was 6.64×10^2 TOPS/W. The common definition of the unit "TOPS/W" represents the tera-operations implemented per second per Watt, i.e., the tera-operations implemented per joule. The energy, or photons and the operations are not necessarily consumed or implemented simultaneously. The certain light energy can transit through multiple layers and implement multiple operations serially, instead of separating for each operation simultaneously. By reusing the photons across layers, the average photon number in each operation is possibly below 1 (*54-56*).

Supplementary Text 2: Energy analysis across different computational precisions, dimensionalities and layer numbers

1) The precision (SNR) at different incident energy levels

We considered both the read-out noise and shot noise in the energy analysis of LightGen, according to the extensively validated noise models (*14, 54*). According to the instrument manual of the sensor we used, i.e., the sCMOS camera (Tucsen Photonics Co., Ltd, Dhyana 400BSI), the photodetector of each detected pixel exhibits a quantum efficiency of η=0.92 at a wavelength of 532 nm and a root mean square (RMS) read-noise of σ_r=1.2e^-. We considered an input image with $N_{px}\times N_{px}$ pixels and LightGen composed of L meta-surface layers, each with a transmittance τ_{meta}, and a fiber array for optical latent space, with a transmittance τ_{FA}. For the incident light with a total energy J_{tot}, the energy per photon is $E_{ph}=hc/\lambda$, where h is the Planck's constant, 6.626×10^{-34}J·s; c is the light speed in vacuum, 3.0×10^8m·s^{-1} and λ is the wavelength of incident light, 532nm. Hence the incident photon number was $N_{ph}=J_{tot}/E_{ph}$. After the propagation through LightGen, the photon number N_{ph}^* reaching the sensor became $N_{ph}^*=\tau_{meta}^L\tau_{FA}N_{ph}$. Then the corresponding number of electrons $N_{\text{e,tot}}$ could be derived from $N_{\text{e,tot}}=\eta N_{\text{ph}}^*$.

According to the energy distribution, the corresponding noise-free number of electrons detected by the $(x, y)th$ photodetector $S(x, y)$, could be derived from

$$S(x,y) = I(x,y)\frac{N_{e,tot}}{\sum_{x,y} I(x,y)} \tag{S31}$$

where $I(x, y)$ was the intensity of the $(x, y)th$ pixel of the incident image. We then considered the following two noises. (i) Shot noise. The detected number of electrons of the signal with shot noise N_{shot} (x, y) was derived from the Poisson distribution Pois($S(x, y)$) according to extensively-validated modeling of shot noise. (ii) Read-out noise. The detected number of electrons of the read-out noise N_{read} (x, y) was derived from the Gaussian distribution $N(0,\sigma_r^2)$. Because the computation in LightGen was all optical and the shot noise appears only when detecting, the number of electrons of the detected noisy signal at the $(x,y)th$ pixel $N_{noisy}(x, y)$ became $N_{noisy}(x, y)=N_{shot}(x, y)+ N_{read}(x, y)$. Therefore, the output SNR by LightGen could be derived from:

$$SNR = 10\log_{10}\left(\frac{\sum_{x,y} S^2(x,y)}{\sum_{x,y}\left[N_{noisy}(x,y) - S(x,y)\right]^2}\right) \tag{S32}$$

Here we displayed two metrics of SNR. $SNR_{LightGen}$ was calculated with noisy output of LightGen and the noise-free output of LightGen, to show the influence of the noise to precision. SNR_{gt} was calculated with the noisy output of LightGen and the ground truth from the electronic digital dataset. Many generative AI tasks, such as semantic image/video generation and style transfer do not have a ground truth from the electronic digital dataset. Therefore, we regarded $SNR_{LightGen}$ as a better demonstration of the relationship between the energy and output precision.

Figure S10A displayed the change of computation precision ($SNR_{LightGen}$ and SNR_{gt}) under varying incident energy by LightGen. The results were calculated in 3D reconstruction tasks because it had ground truths from electronic digital datasets to calculate SNR_{gt}. LightGen was composed of six meta-surfaces. The output precision ($SNR_{LightGen}$) increased along with the energy of incident light, corresponding well with theoretical models (*57*). The star marked the experimental energy of incident light we used to calculate the experimental energy efficiency of LightGen and demonstrated the experimental task performances of large-scale generative AI by LightGen in Fig. 3-4. It showed that even using a relatively high incident energy for adequate calculation precision, LightGen maintained 2 orders of magnitude higher experimental energy efficiency (6.64×10^2 TOPS/W) than NVIDIA A100 (1.04 TOPS/W).

2) The energy for fixed precision (SNR) at different layer numbers of meta-surfaces

Figure S10B showed the change of incident energy under different layer numbers of the meta-surfaces in LightGen, to achieve a fixed adequate computation precision ($SNR_{LightGen}$ and SNR_{gt}). The SiO_2 meta-surfaces we used with different etching depths provided transmission energy loss. We experimentally measured the average transmission coefficient τ_{meta} of the meta-surfaces in LightGen, which was 0.889. It was relatively lower than the theoretical transmission coefficient 0.932 of the SiO_2 material JGS1 we used for the meta-surfaces, because of the patterns on the meta-surfaces (*58, 59*).

We showed the incident light energy to achieve fixed SNRs (output precision) under various numbers of meta-surface layers in Fig. S10B. The results were calculated in 3D reconstruction tasks. The required incident energy increased along with the number of meta-surface layers exponentially. It also corresponded well with other models (*60*). The required energy (below 10^{-11}J) also indicated that we used adequate incident energy in Fig. 3-4 and the calculation of energy efficiency of LightGen.

3) The energy for fixed precision (SNR) at different input dimensionalities

Figure S10C displayed the change of incident energy under different input dimensionalities of LightGen, to achieve a fixed adequate output precision ($SNR_{LightGen}$ and SNR_{gt}). The results were calculated in 3D reconstruction tasks with six meta-surfaces. The required energy increased when the dimensionality, i.e., the number of 3 μm pixels of the meta-surfaces, increased. It corresponded well with other models (*61*). The required energy (below 10^{-11}J) also indicated that we used adequate incident energy in Fig. 3-4 and the calculation of energy efficiency of LightGen.

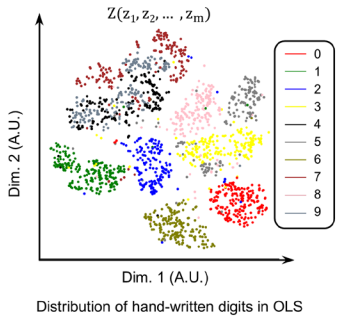


Figure S1. An example of the distribution of 10-category handwritten digits in OLS. The dataset was from MNIST (Modified National Institute of Standards and Technology). The data from different categories were unsupervisedly learned to cluster in different areas in OLS, indicating that OLS designed based on characteristics of light field effectively learned the semantic meanings of input data.

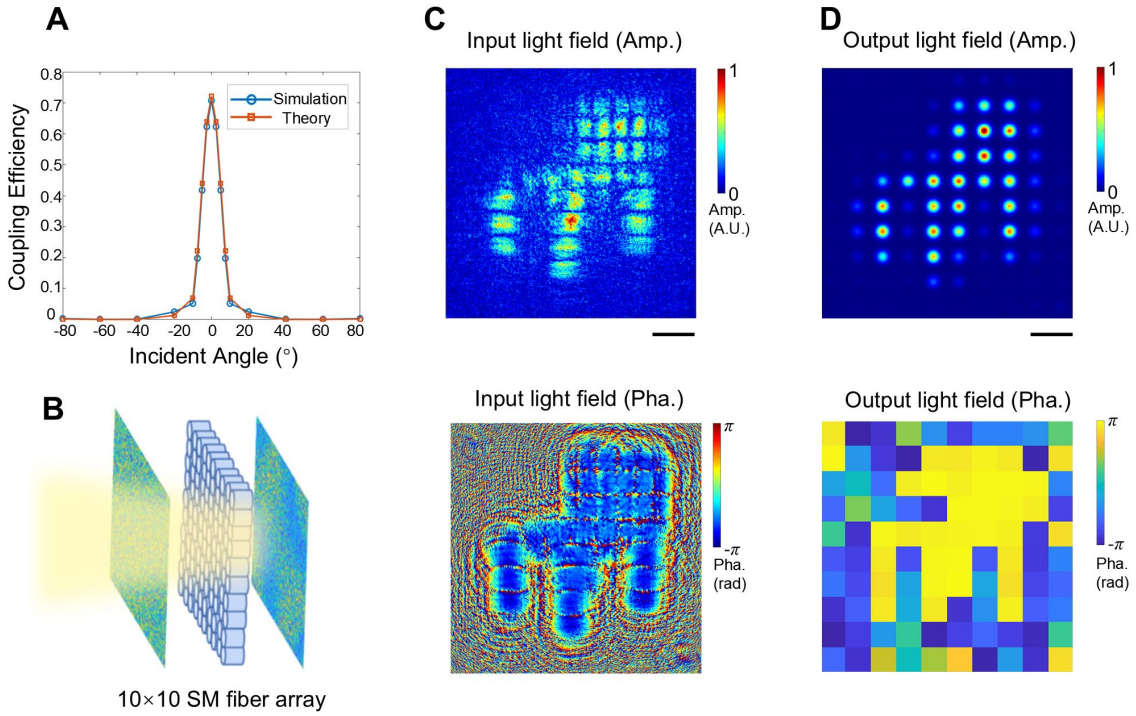


Figure S2. Coupling of OLS. (**A**) Analysis of the coupling efficiency of both theoretical and FDTD simulations for different incident angles. The complex light field can be decomposed into the addition of light from diverse incident angles. (**B**) A schematic diagram of the input optical field propagating through a 10×10 OLS. (**C**-**D**) Simulated distribution of the input and output optical field of OLS, including the amplitude and phase, demonstrating the transformation of the optical field by OLS. Scale bar: 200 μm. Amp.: amplitude. Pha.: phase.

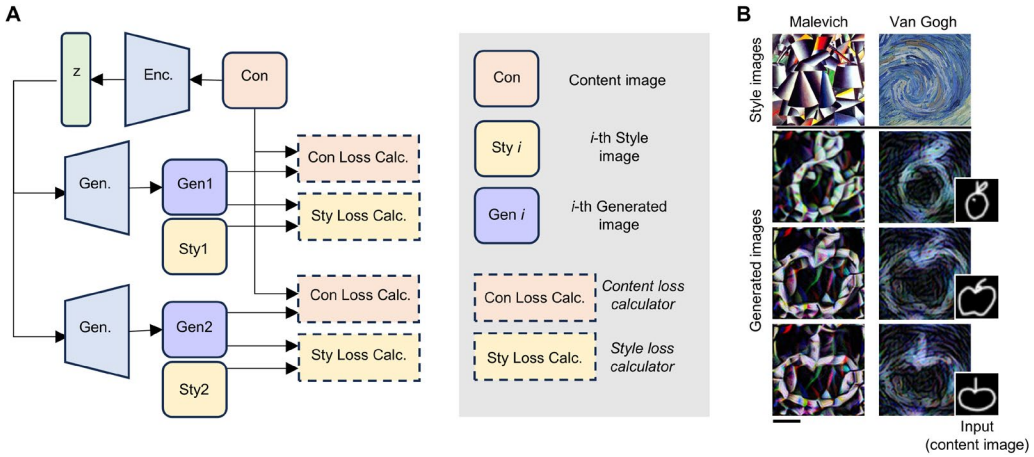


Figure S3. Reconfiguration of LightGen. (**A**) The algorithm to reconfigure LightGen with a joint encoder and switchable generators for style transfer. (**B**) The schematic of the style transfer performance of LightGen using a constant joint encoder and diverse generators corresponding to different styles. Scalebar: 100 μm.

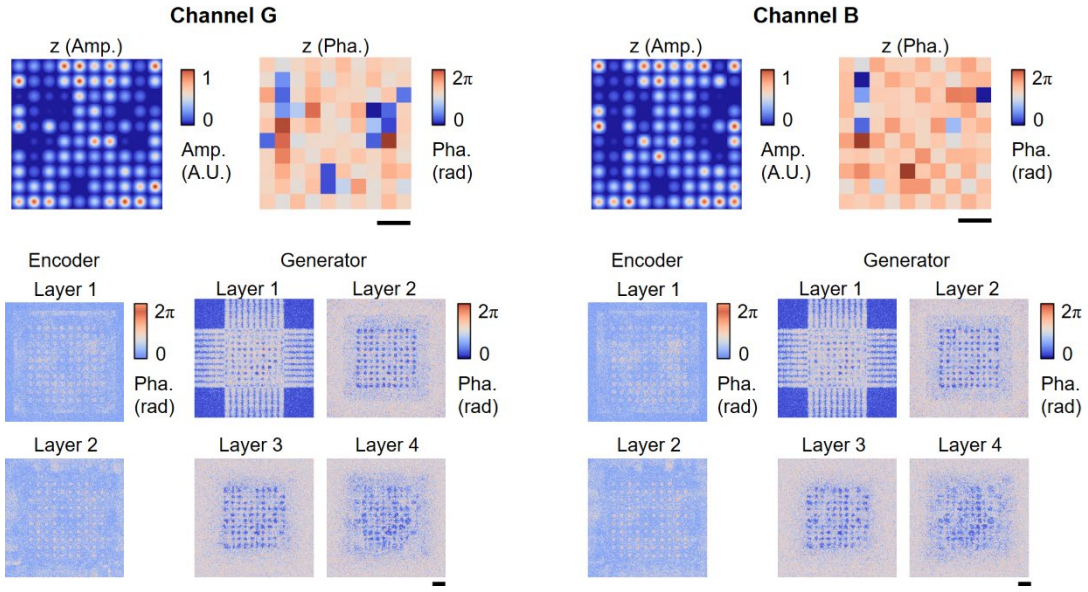


Figure S4. Examples of OLS and optical weights of LightGen. An example of the optical weights for encoder and generator respectively, and the amplitude and phase of the established OLS, corresponding to the G and B channels of the framed image in Fig. 2C. The R channel is shown in Fig. 2A-B. Scalebar: 200 μm.

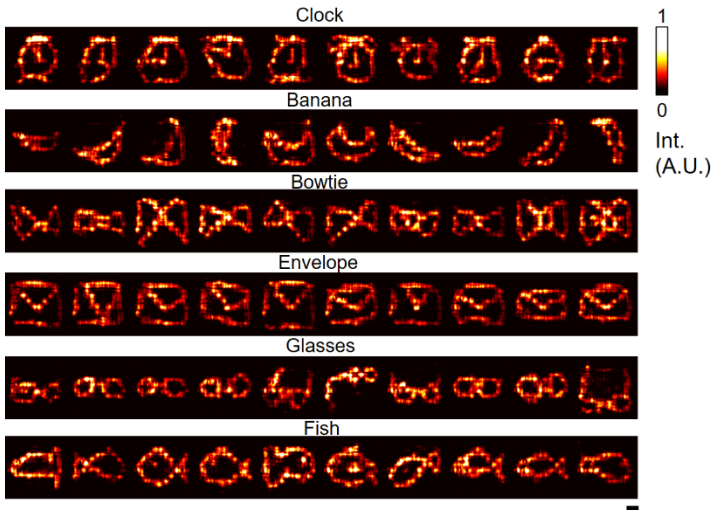


Figure S5. Experimentally generated items with different species and features by LightGen. We showed sixty images experimentally generated by LightGen from six species (clock, banana, bowtie, envelope, glasses and fish). Each image had different, even contrary details and styles but remained to belong to the same category. For example, circular, elliptic and tortuous clocks with different displayed times (the first row); bananas with various directions, thickness and curvatures (the second row); fish of multifarious fins, tails and postures (the sixth row). It showed that LightGen experimentally encoded features of both categories and details, and generated as well as manipulated them respectively. Scalebar: 200 μm.

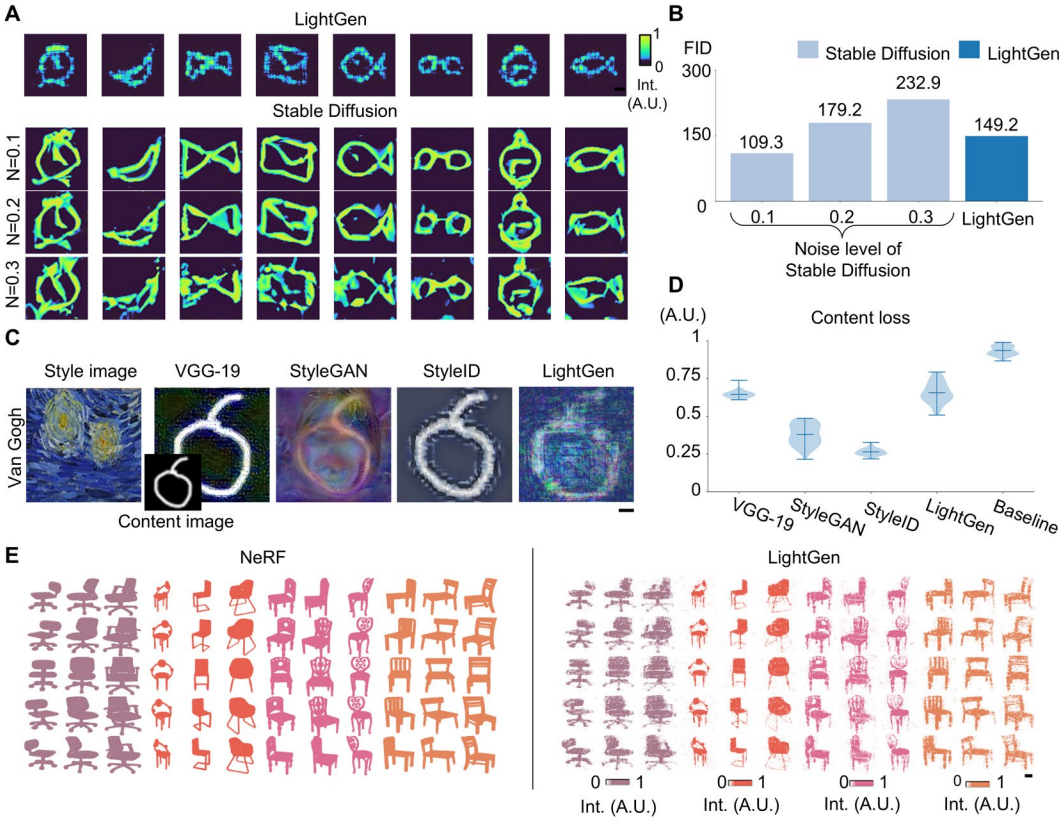


Figure S6. Experimental comparison of LightGen, Stable Diffusion, StyleGAN, and NeRF in diverse generative AI tasks. (**A**) Experimental semantic generation by LightGen and Stable Diffusion on Quick, Draw! Dataset (*31)*. N: Noise level. Scalebar: 200 μm. (**B**) FID of experimental LightGen generation compared with Stable Diffusion on Quick, Draw! Dataset. We noted that the FID can be lower in both LightGen and Stable Diffusion if using larger datasets according to the characteristics of FID (*62*). (**C-D**) Experimental style transfer performance of LightGen, compared with VGG-19, StyleGAN and StyleID. The content loss showed the extent of retention of the content images. 'Baseline' was calculated between the style images and content images to serve as a baseline to eliminate the influence of style-image-content-image similarity. Error bars in the violin plots denote the data range (minimum to maximum), with the central lines representing the average values. Scalebar: 200 μm. (**E**) Experimental 3D reconstruction performance of LightGen and NeRF (*38)*. Scalebar: 200 μm.

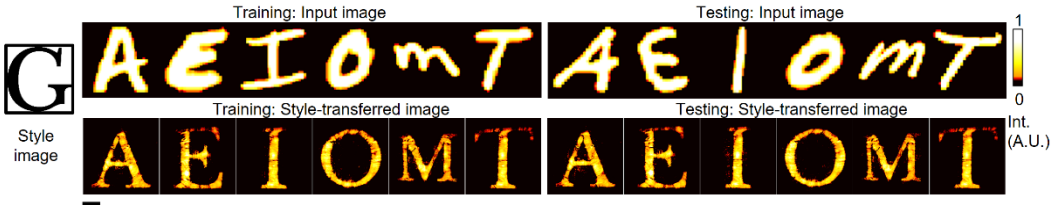


Figure S7. The experimental style transfer of LightGen for letters on both training and testing datasets. LightGen achieved good performance on both training and testing datasets, showing its capability of generalization. It was trained on the EMNIST Dataset. Scalebar: 200 μm.

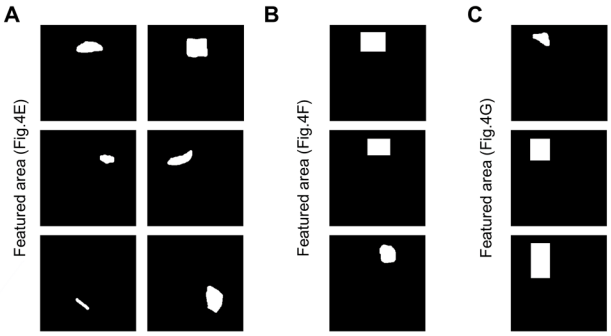


Figure S8. Featured areas in the quantitative analysis of semantic 3D generation. We labeled the areas of goal features to manipulate to evaluate the experimental performance of LightGen for Fig. 4E **(A)**, Fig. 4F **(B)** and Fig. 4G **(C).**

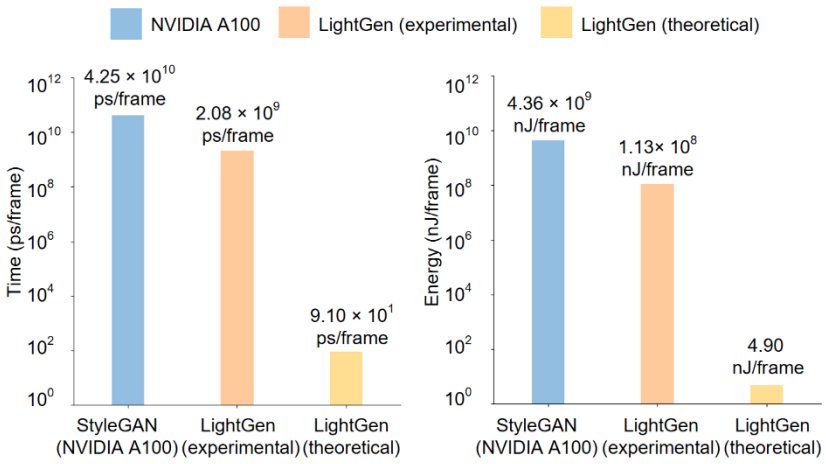


Figure S9. Experimental systemic time and energy consumption of LightGen and the state-of-the-art electronic network when achieving comparable output performance. We experimentally tested digital StyleGAN on NVIDIA A100 and the results were averaged over 450 times of single forward inferences. The measured results of electronic networks corresponded well with published reports (*53*). The results were quite similar under tests on different input and style images. The measurement of experimental and theoretical indices of LightGen are listed in detail in Supplementary Text 1.

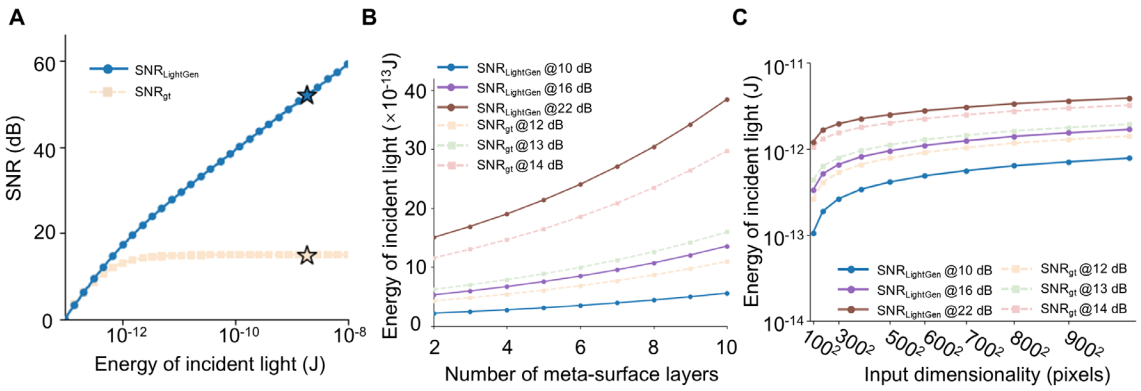


Figure S10. Energy analysis of LightGen considering output precision, number of meta-surface layers and input dimensionality. (**A**) Output precision with varying incident light energy. The star marked the incident light energy used in Fig. 4. SNR: signal-to-noise ratio. $SNR_{LightGen}$ was calculated with noisy output of LightGen and the noise-free output of LightGen, to show the influence of the noise to precision. SNR_{gt} was calculated with the noisy output of LightGen and the ground truth from the electronic digital dataset. (**B**) The incident light energy for different numbers of meta-surface layers at fixed SNRs. (**C**) The incident light energy for different input dimensionalities, i.e., the number of 3 μm pixels, at fixed SNRs.

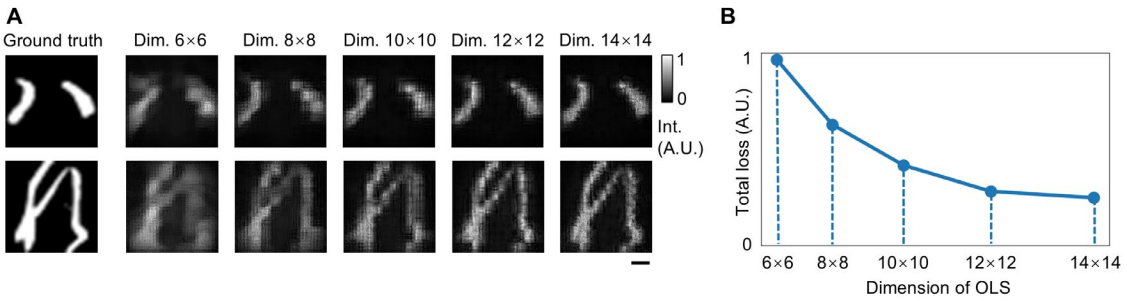


Figure S11. The performance of LightGen with different OLS dimensions. (**A**) The output of LightGen with the dimensions of OLS as 6×6, 8×8, 10×10, 12×12 and 14×14, respectively. Dim.: dimension of OLS. It is demonstrated on the K-MNIST Dataset. Scalebar: 200 μm. (**B**) The total loss of LightGen with different OLS dimensions.

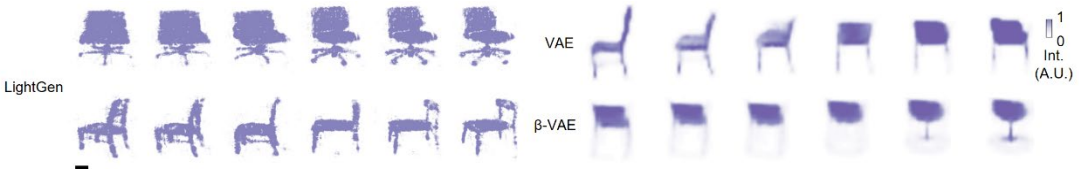


Figure S12. Experimental comparison of LightGen with digital VAE and digital *β*-VAE in visual generation and semantic manipulation. The output results of digital VAE and *β*-VAE are cited from (*40*). Scalebar: 200 μm.

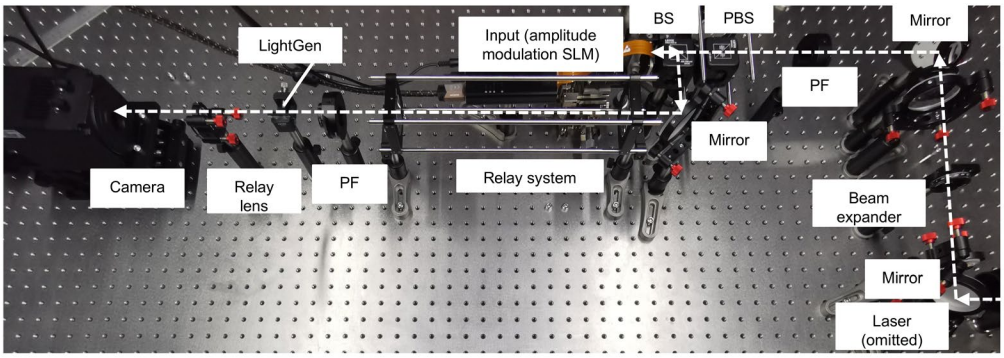


Figure S13. Experimental set-up to test LightGen. It is a wide-angle shot to include the set-up. The laser was a single-mode 532nm laser (Changchun New Industries Optoelectronics Tech. Co., Ltd, MSL-U-532-300mW). PF: polarizing film. PBS: polarizing beam splitter. BS: beam splitter. SLM: spatial light modulator (for input).

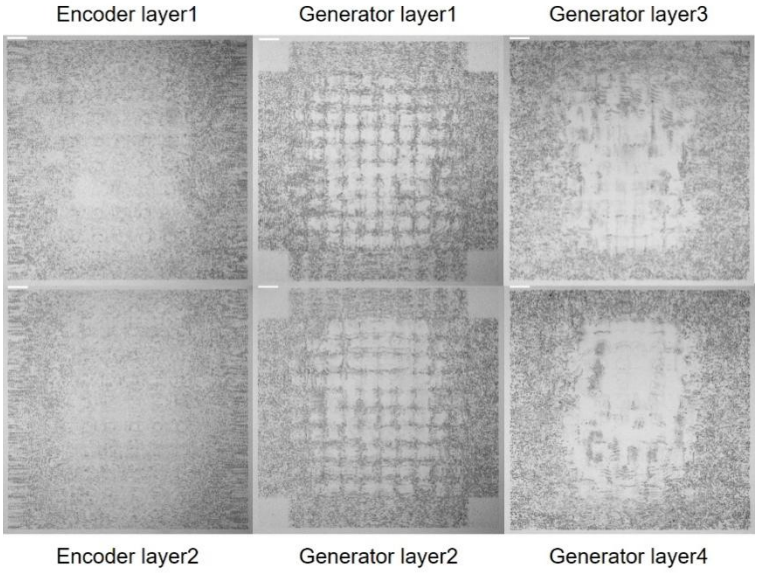


Figure S14. Microscopy photos of meta-surfaces in LightGen. The photonic weights were overlay photolithographed by 8 levels. Scalebar: 100 μm.

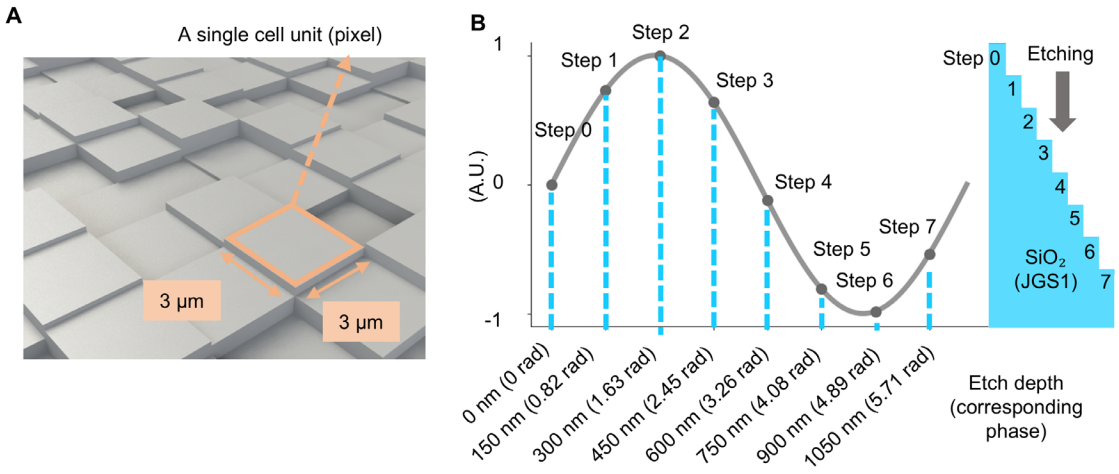


Figure S15. The unit cell schematic of LightGen. (**A**) The unit cell schematic for 3μm pixels in meta-surfaces of LightGen. (**B**) The mapping relationship of modulation phases and steps of etching depths in meta-surfaces.

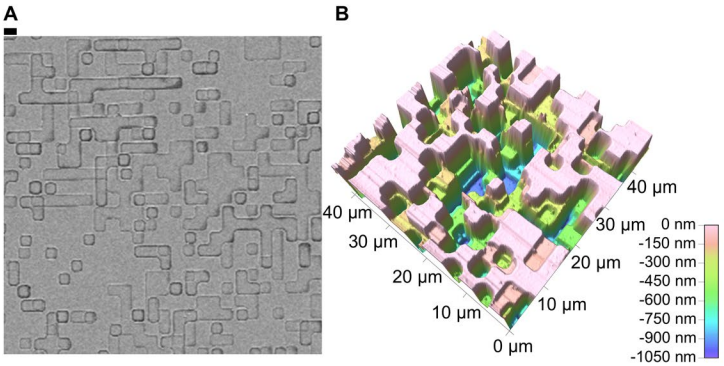


Figure S16. High-resolution SEM and AFM images of meta-surfaces in LightGen. (**A-B**) Examples of the measured high-resolution SEM and AFM photos showing the measured detailed features of the 3µm pixels. Scalebar of SEM: 3 µm. Scalebar of AFM is labeled in the figure.

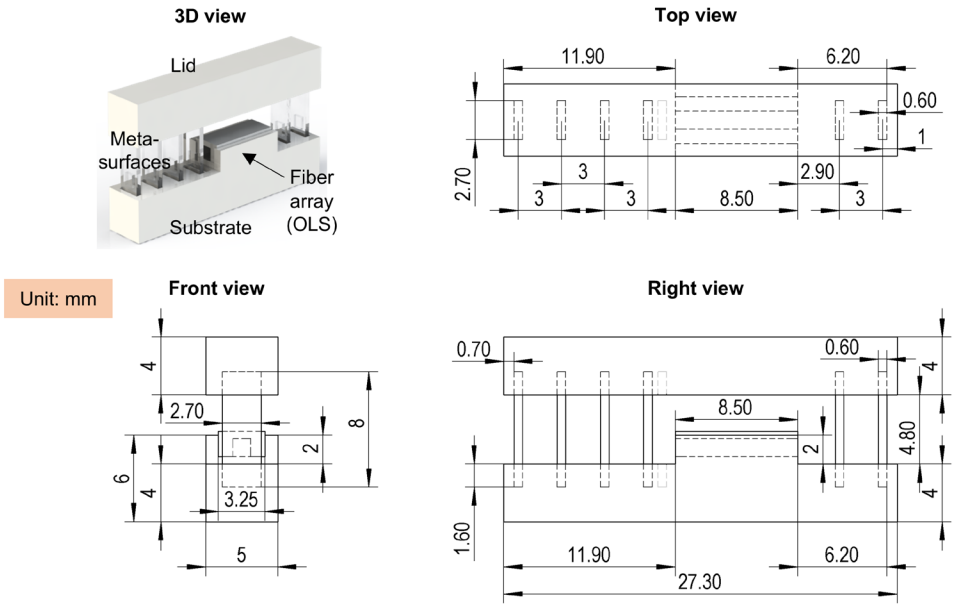


Figure S17. The 3D-printed holder and packaged LightGen. Unit: mm.

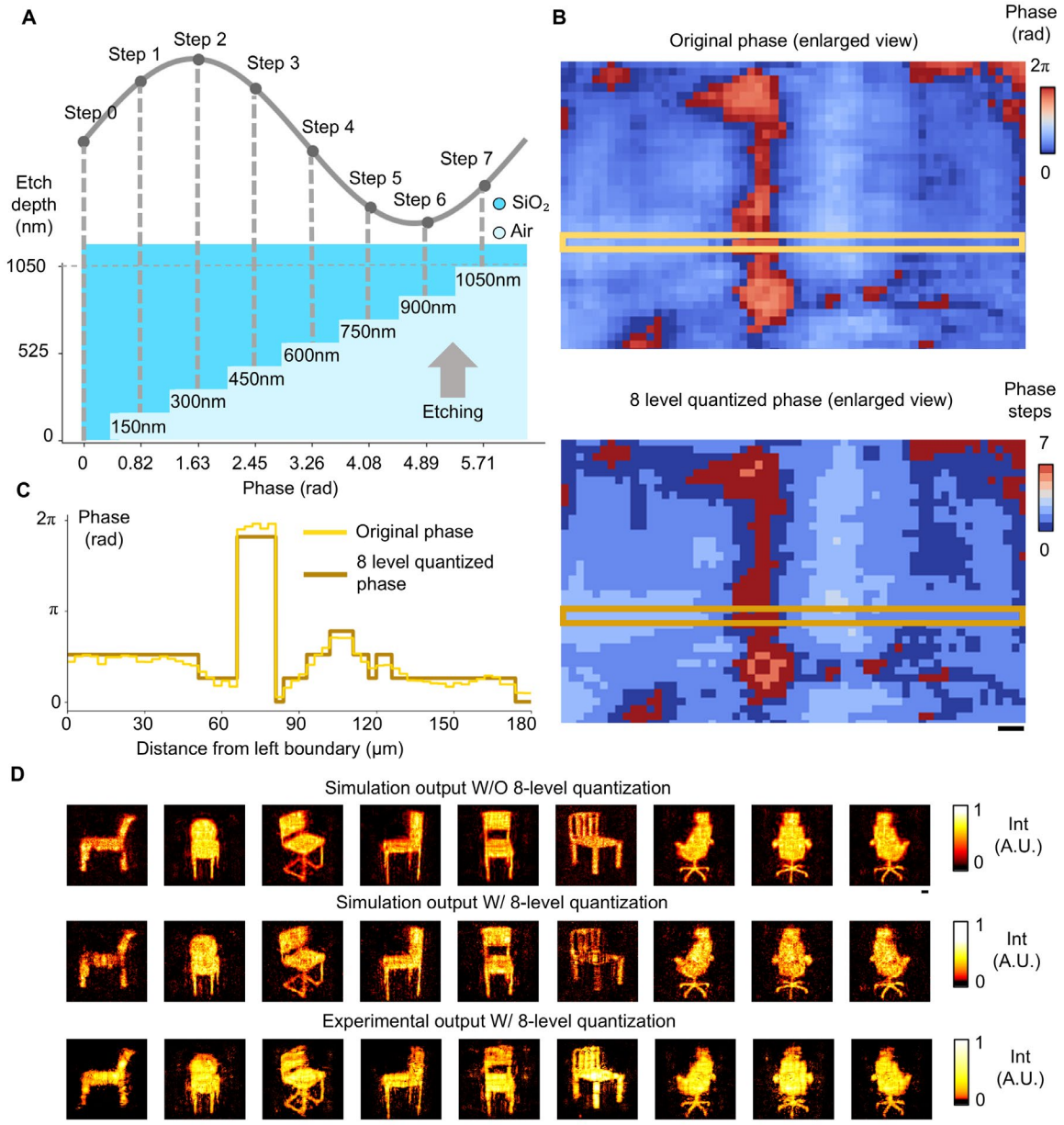


Figure S18. The quantization method and experimental results of LightGen. (**A**) The detailed quantization method and corresponding relationships used in LightGen. (**B**) An example of the phase weights in the meta-surface before and after quantization. Scalebar: 10 μm. (**C**) Zoom-in comparison of the phase weights in the framed area in (**B**). (**D**)The output comparison of simulation W/O, simulation W/ and experimental W/ 8-level quantization in all meta-surfaces of LightGen. Scalebar: 100 μm.

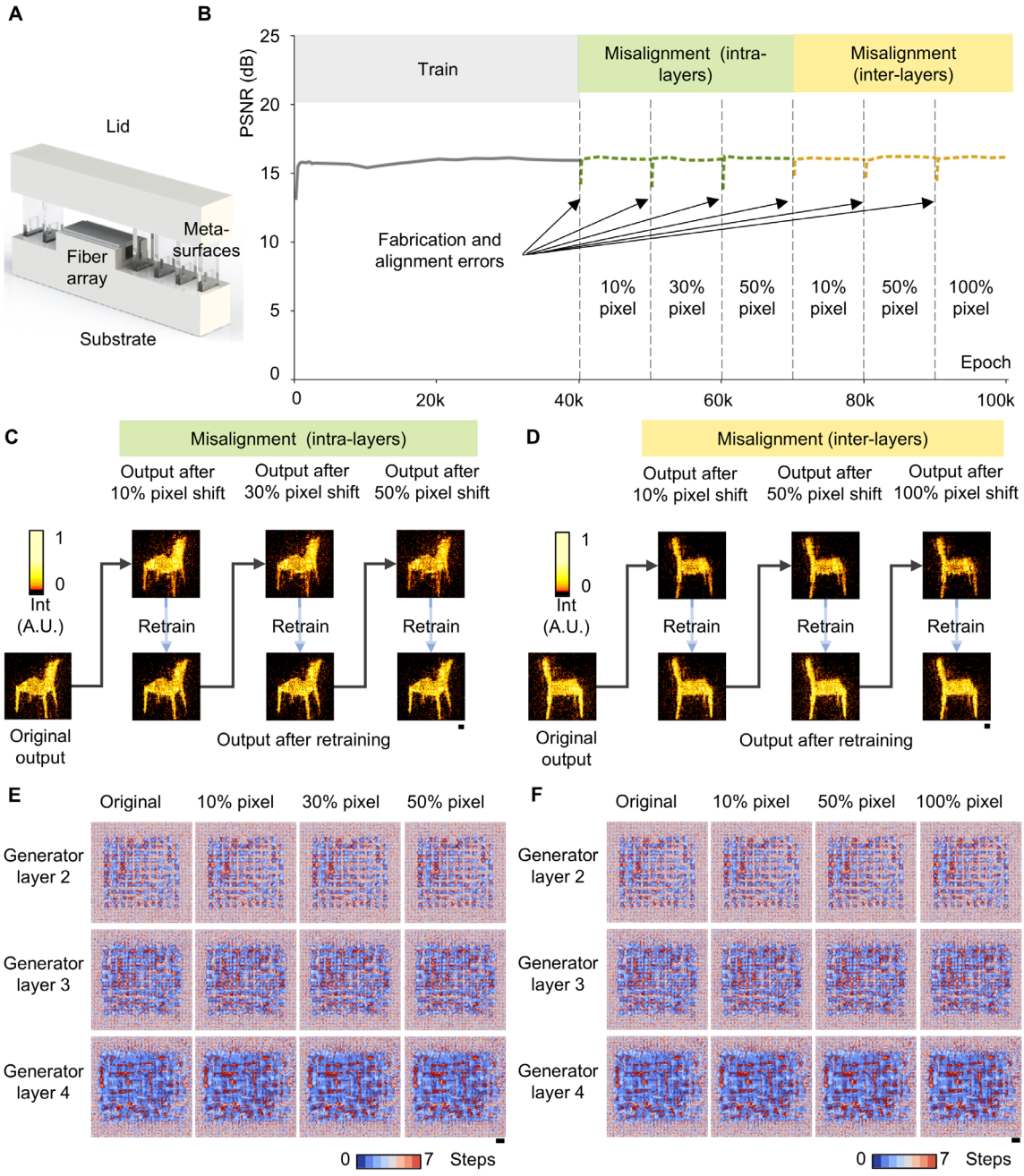


Figure S19. Alignment and calibration of LightGen. (**A**) A schematic diagram to show the packaging and alignment of LightGen. (**B**) PSNR of images generated by LightGen after misalignment and calibration. (**C**) Example images to illustrate the qualitative performance of LightGen after intra-layer misalignment and calibration. The etch layers in generator layer 1 were shifted in opposite directions by 10% pixel (0.3 μm), 30% pixel (0.9 μm) and 50% pixel (1.5 μm) respectively as misalignment. Scalebar: 100 μm. (**D**) Example images to illustrate the qualitative performance of LightGen after misalignment between different meta-surfaces and calibration. The generator layer 1 was shifted horizontally by 10% pixel (0.3 μm), 50% pixel (1.5 μm) or 100% pixel (3.0 μm) respectively as

misalignment. Scalebar: 100 μm. (**E-F**) The weights of meta-surfaces after calibration for different intra- and inter-layer misalignments. Scalebar: 100 μm.

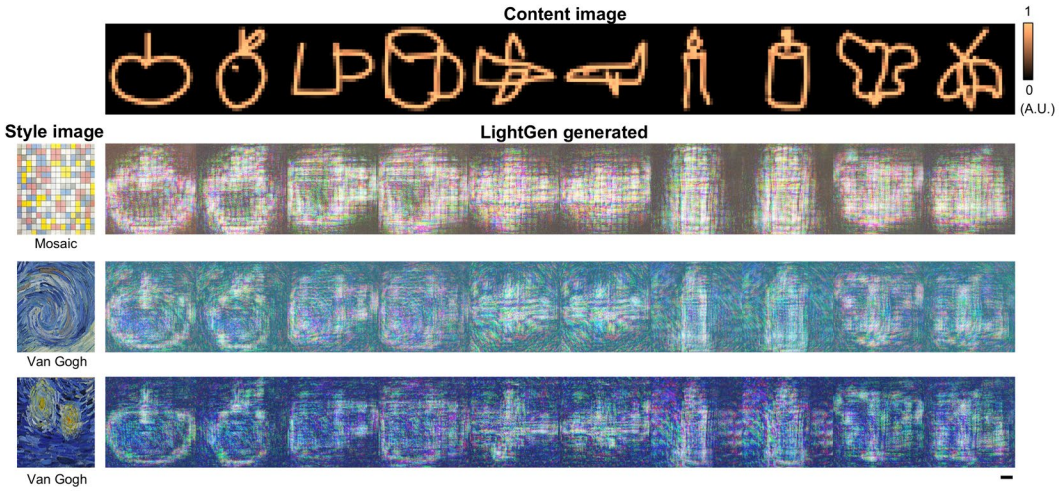


Figure S20. Experimental results of style transfer with LightGen. We showed ten examples of each style to show that LightGen transferred the styles as well as retained the contents during intelligent computing on various styles, e.g., Mosaic and Van Gogh styles. Scalebar: 200 μm.

Reference	Method	Capable of vision generative AI tasks? (input resolution)	Color mode	Computing density (TOPS/mm^2)	Computing Speed (TOPS)	Neuron number	Energy Efficiency (TOPS/W)
Diffusion Models (*63*)	Digital electronic	Yes (256×256)	RGB	0.38 (NVIDIA A100)	312 (NVIDIA A100)	-	1.04 (NVIDIA A100)
ViT-GAN (*64*)	Digital electronic	Yes (256×256)	RGB	0.38 (NVIDIA A100)	312 (NVIDIA A100)	-	1.04 (NVIDIA A100)
2D- material sensor (***Nature***, 2020) (*65*)	Opto-electronic	Yes (3×3)	Gray	-	0.001	27	-
Optical GAN (***Science Advances***, 2022) (*66*)	Opto-electronic	Yes (14×14)	Gray	0.0000006 (optical part)	0.000000004 (optical part)	-	0.005
Taichi (***Science***, 2024) (*19*)	Optical	Yes (8×8, chiplet)	RGB	1.8×10^3	-	4.3×10^3	160.82
Photonic tensor core (***Nature***, 2025) (*67*)	Opto-electronic	-	RGB	0.047	65.5	1.6×10^4 (single PTC)	0.81
PACE (***Nature***, 2025) (*42*)	Opto-electronic	No	-	-	8.19	4.1×10^3	2.38
LightGen (experimental)	Optical	**512×512**	**RGB**	**2.62×10^2**	**3.57×10^4**	**2.2×10^6**	**6.64×10^2**
LightGen (theoretical)	Optical	**512×512**	**RGB**	**4.17×10^7**	**5.69×10^9**	**2.2×10^6**	**1.06×10^8**

Table S1. Experimental performance comparison with state-of-the-art electronic and photonic chips in generation. We compared LightGen with other digital and photonic synthesis chips in terms of systemic input resolution, color mode, computing density, computing speed, and energy efficiency. The diffusion model and Vision Transformer GAN (ViT-GAN) were evaluated based on computational results obtained with NVIDIA A100 (*6*). Its referenced computing density, computing speed, and energy efficiency data were sourced from the official website. The researches listed here were mainly integrated chips, so some benchtop systems were not included. '-' means not mentioned in the reference.

Indices	LightGen	Diffractive neural network (*Science*, 2018) (*23*)	MZI-based chip (*Nature,* 2025) (*42*)	MRR-based chip (*Nature,* 2022) (*15*)
Packaging method	3D	Unpackaged discrete desktop system	2.5D	2D
Scale of photonic neuron number	2.16×10^6	10^5	10^3	66

Table S2. Comparison of example photonic-computing packaging schemes. LightGen achieved large-scale photonic neurons and integration simultaneously. MZI: Mach–Zehnder Interferometer. MRR: Micro-Ring Resonator.

Symbol	Parameter	Measured results (case 1)	Measured results (case 2)
Num_e	The neuron number in encoder	400×400×2	600×600×2
Num_g	The neuron number in generator	400×400×4	600×600×4
N_{op}	Operation number	1.02×10^{11}	5.18×10^{11}
A	Footprint (mm^2)	136.5	136.5
P_o	Average power of the incident light (μW)	249.3	249.3
$P_{detection}$	Power of the camera (W)	49.29	49.29
P_{Input}	Power of the SLM to input images (W)	4.56	4.56
E_{sys}	Energy (nJ)	4.90	4.90
t_{frame}	Theoretical computing time for one frame (s)	9.1×10^{-11}	9.1×10^{-11}
S(experimental)	Experimental computing speed (TOPS)	7.07×10^{3}	3.57×10^{4}
S(theoretical)	Theoretical computing speed (TOPS)	1.12×10^{9}	5.69×10^{9}
Energy $efficiency_{sys}$ (experimental)	Experimental energy efficiency (TOPS/W)	1.31×10^{2}	6.64×10^{2}
Energy $efficiency_{sys}$ (theoretical)	Theoretical energy efficiency (TOPS/W)	2.08×10^{7}	1.06×10^{8}
D(experimental)	Experimental computing density ($TOPS/mm^2$)	51.76	2.62×10^{2}
D(theoretical)	Theoretical computing density ($TOPS/mm^2$)	8.21×10^{6}	4.17×10^{7}

Table S3. Calculation of the experimental performance of LightGen. The two cases indicated the LightGen with different pixel numbers in each layer. The detailed calculation is listed in Supplementary Text 1.

Components of LightGen	Footprint (length × width = footprint)
Encoder	6.2mm×2.7mm = 16.74 mm^2
Fiber array (OLS)	8.5mm× 3.25mm = 27.625 mm^2
Generator	11.9mm× 2.7mm = 32.13 mm^2
The 3D-printed holder (including all above components)	27.3mm× 5mm = 136.5 mm^2
Total of LightGen	136.5 mm^2

Table S4. Footprints of all components in LightGen. The table lists the footprints of all components in LightGen. All components were packaged within the area of the 3D-printed holder.

Movies S1. Video experimentally generated by LightGen with adjustable features.

References and Notes

1. L. A. Abriata, The Nobel Prize in Chemistry: Past, present, and future of AI in biology. *Commun. Biol.* **7**, 1409 (2024).

2. X. L. Meng, AI has won Nobel Prizes in hard science: Can humans be smarter and softer on each other? *Harv. Data Sci. Rev.* **6**, 4 (2024).

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